

Lab 1 Design a VPC and Associated Subnets

In this challenge, you will learn how to create a Virtual Private Cloud (VPC) and private subnets. Note: Once you begin a challenge you will not be able to pause, save, or return to your progress. Please ensure you have set aside enough time to complete the challenge before you start.

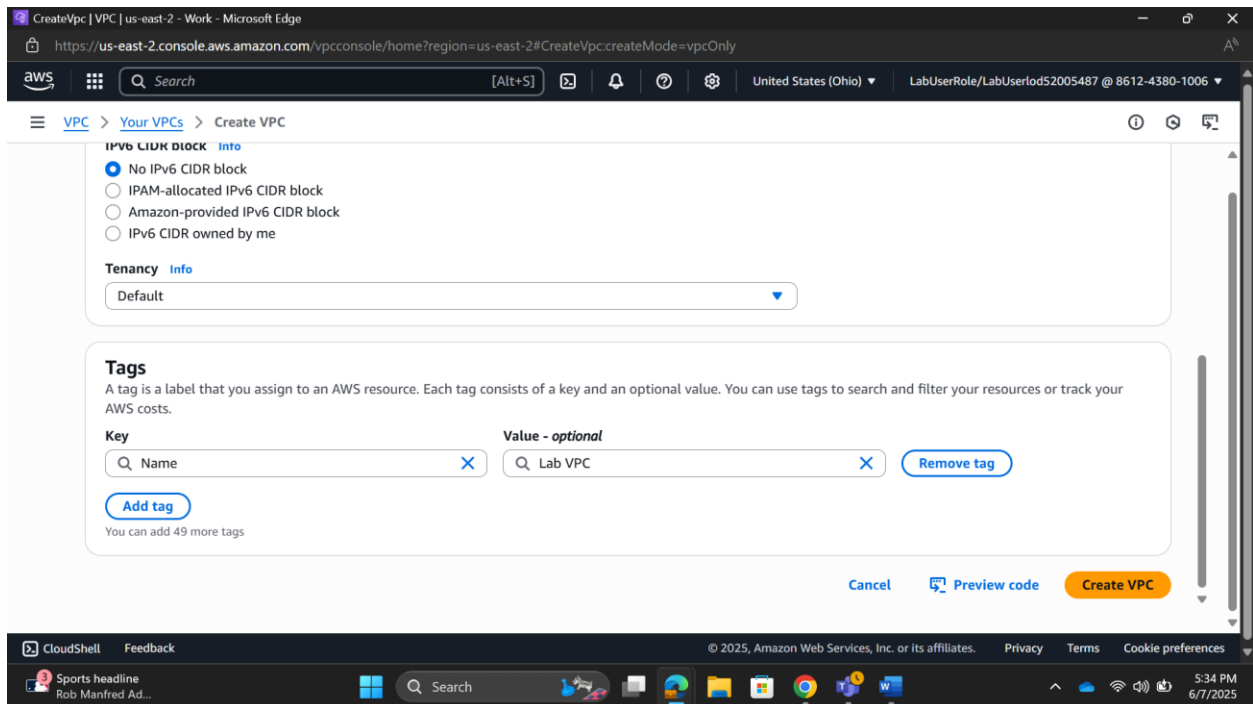
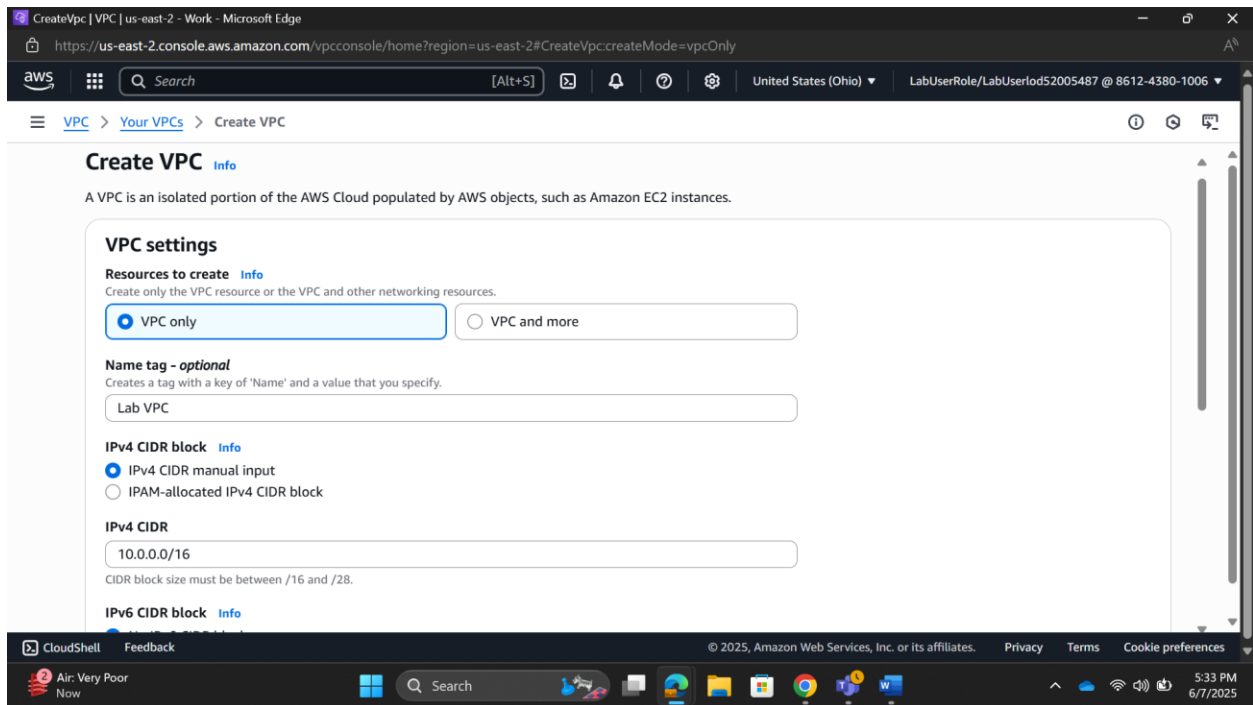
Task 1: Create a new VPC

VPC -> Virtual private cloud -> Your VPCs -> Create VPC

The screenshot shows the AWS Management Console interface for the 'us-east-2' region. The main content area is titled 'Your VPCs (1)' and contains a table with the following data:

Name	VPC ID	State	Block Public...	IPv4 CIDR
-	vpc-09e060e894bdfdf592	Available	Off	172.31.0.0/16

The left sidebar shows the 'Virtual private cloud' section expanded, with 'Your VPCs' selected. The top navigation bar shows the user is in the 'us-east-2' region. The bottom status bar shows the time as 5:32 PM on 6/7/2025.



Click Create VPC

aws | vpcs | VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#vpcs:

Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52005487 @ 8612-4380-1006

VPC > Your VPCs

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Your VPCs (2) Info

Find VPCs by attribute or tag

Last updated less than a minute ago

Actions Create VPC

Name	VPC ID	State	Block Public...	IPv4 CIDR
-	vpc-09e060e894bdfd592	Available	Off	172.31.0.0/16
Lab VPC	vpc-071740c25d0bdcfb2	Available	Off	10.0.0.0/16

Select a VPC above

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Sports headline Rob Manfred Ad...

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Task 2: Create Private Subnets

VPC -> Virtual private cloud -> Subnets

aws | subnets | VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#subnets:

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VPC > Subnets

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

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NAT gateways

Peering connections

Route servers

Subnets (3) Info

Find subnets by attribute or tag

Last updated 4 minutes ago

Actions Create subnet

Name	Subnet ID	State	VPC
-	subnet-0d4d5c349eb711804	Available	vpc-09e060e894bdfd592
-	subnet-080d47ca0e5a1c645	Available	vpc-09e060e894bdfd592
-	subnet-0fdd6b156deab2c32	Available	vpc-09e060e894bdfd592

Select a subnet

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VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpccconsole/home?region=us-east-2#CreateSubnet:

aws

Search

[Alt+S]

United States (Ohio)

LabUserRole/LabUserIod52005487 @ 8612-4380-1006

VPC > Subnets > Create subnet

Create subnet Info

VPC

VPC ID

Create subnets in this VPC.

vpc-071740c25d0bdcfb2 (Lab VPC)

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

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VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpccconsole/home?region=us-east-2#CreateSubnet:

aws

Search

[Alt+S]

United States (Ohio)

LabUserRole/LabUserIod52005487 @ 8612-4380-1006

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private Subnet 1

The name can be up to 256 characters long.

Availability Zone Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (Ohio) / us-east-2a

IPv4 VPC CIDR block Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

256 IPs

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VPC | us-east-2 - Work - Microsoft Edge
https://us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#CreateSubnet:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52005487 @ 8612-4380-1006

VPC > Subnets > Create subnet

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24 256 IPs

▼ **Tags - optional**

Key	Value - optional	
Q Name	Q Private Subnet 1	Remove

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

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Create subnet

Same way will create private subnet 2 and private subnet 3

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https://us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#CreateSubnet:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52005487 @ 8612-4380-1006

VPC > Subnets > Create subnet

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private Subnet 2
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (Ohio) / us-east-2b

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24 256 IPs

▼ **Tags - optional**

Key	Value - optional	
Q Name	Q Private Subnet 2	Remove

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

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VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#CreateSubnet:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52005487 @ 8612-4380-1006

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private Subnet 3
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (Ohio) / us-east-2c

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.3.0/24 256 IPs

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subnets | VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#subnets:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52005487 @ 8612-4380-1006

VPC > Subnets

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

You have successfully created 1 subnet: subnet-0ecb86dd75daac766

Last updated less than a minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC
☐	-	subnet-080d47ca0e5a1c645	Available	vpc-09e060e894bdfd592
☐	-	subnet-0fdd6b156deab2c32	Available	vpc-09e060e894bdfd592
☐	Private Subnet 1	subnet-08eb0f7f1e105d8cc	Available	vpc-071740c25d0bdcfb2 Lab
☐	Private Subnet 2	subnet-01c7cb956088647b9	Available	vpc-071740c25d0bdcfb2 Lab
☐	Private Subnet 3	subnet-0ecb86dd75daac766	Available	vpc-071740c25d0bdcfb2 Lab

Select a subnet

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labclient.labondemand.com/Instructions/ExamResult/36b7f03b-c7ff-41cd-bc34-469eefc15f0f - Work - Microsoft Edge

https://labclient.labondemand.com/Instructions/ExamResult/36b7f03b-c7ff-41cd-bc34-469eefc15f0f

Congratulations!

Print

Clear out some room in the trophy case!



100%

Areas where you excelled

- **Create a new VPC**
You completed this area with great success.
- **Create Private Subnets**
You showed your skills in this area and finished with success.

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You are done with the lab!!!!